Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/06/2005
PI Name:	Szeri, Andrew J. Ph.D.		
Project Title:	Using Guided Ultrasound to Control Bubbles in Reduced Gravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	01/01/2005	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Matula, Thomas (University of Washington)		
Grant/Contract No.:	NNC05GA44G		
Performance Goal No.:			
Performance Goal Text:			

The goal of this project is to develop techniques and prototypes devices to handle liquids in reduced gravity. These efforts will be focused on the use of ultrasound to control the placement and motion of free and surface-attached bubbles in the liquid.
In reduced gravity, the tendency of the bubbles to rise out of suspension is diminished. This can be problematical because:(i)fluid handling with pumps,etc. may be unreliable if there are bubbles in the liquid (ii) dissolution of gases may be undesirable, as in bioreactors, and (iii) storage volume of the liquid is increased. Finally,(iv) bubbles could also be problematical in materials processing, for example their presence in a weld or crystal could be catastrophic. Bubbles are more likely to stick to surfaces in reduced gravity. This is important when the surfaces of heaters, and the lingering bubbles(v)can reduce heat transfer, and (vi0 can promote unstable flm boiling. It is also important when the surfaces are the surfaces of electrodes in a fuel cell, and the lingering bubbles(vii)reduce the area of the electrode in contact with the

Task Description:	liquid-and hence the efficiency of the device.Finnaly(viii)cells in bioreactors can stick to the surfaces of bubbles, and be liable to nutrient disruption or cavitation damage. The problems associated with bubbles in liquids has been of pressing interest to NASA for decades. experiments on low gravity boiling and on the consequences of bubble inclusions in material processing in low gravity hve been investigated for more than 30 years, culminating in recent experiments on the International Space Station . Ultrasound (US) has been used to cause convective motions in a liquid directly through acoustic streaming. US can also drive the motions of bubbles directly through the Bjerknes force. Secondary Bjerknes forces have been shown to cause agglomeration of smaller bubbles into larger bubbles that may be more easy to handle. Acoustic streaming could be used as a stirring mechansim to detacj bubbles from active surfaces. Surfactants can be exploited to alter the surface tension to enchance or alter these effects. All of the phenomena fall under the scope of this project.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	This project began during FY 2005. The first progress report is due during FY 2006.
Bibliography Type:	Description: (Last Updated:)